
1. Executive Summary

This study analyzes the relationship between **demographic characteristics, household structure, and the geographic distribution of churches** surrounding a selected location in Columbus, Georgia.

Demographic data were aggregated across multiple radii ranging from **1 to 9 miles** from the study location. Church locations were extracted from OpenStreetMap and counted within each radius.

The analysis examined relationships between:

- Median household income
- Educational attainment
- Median age
- Homeownership rates
- Household structure (two-parent vs. single-parent households)
- Church density

A total of **121 churches** were identified within a **9-mile radius** of the study location.

Initial findings indicate that **church presence correlates most strongly with indicators of neighborhood stability**, particularly homeownership rates, household income, and age demographics. Household structure also reveals clear geographic trends, with areas further from the city center showing higher proportions of two-parent households.

While church count increases with geographic radius, **church density per capita decreases slightly as population grows faster than the number of churches**.

Overall, the results suggest that churches tend to be embedded in **stable residential environments**, while areas closer to the urban core exhibit higher concentrations of single-parent households and lower homeownership rates.

2. Research Question

The central research question guiding this study was:

How do demographic and household structure characteristics correlate with the geographic distribution of churches within an urban environment?

Particular attention was given to whether **family structure variables**, such as the proportion of two-parent households versus single-parent households, show observable relationships with church presence.

3. Methodology

Study Location

The analysis was centered on the following address:

**718 8th St
Columbus, Georgia**

Demographic statistics were calculated across expanding geographic radii.

Radius Description

1 mile Immediate neighborhood

3 miles Local community

5 miles Urban residential area

7 miles Extended urban area

9 miles Metropolitan area

Census tract demographic data were spatially aggregated within each radius using area-weighted estimates. Church locations were obtained from OpenStreetMap and counted within each radius.

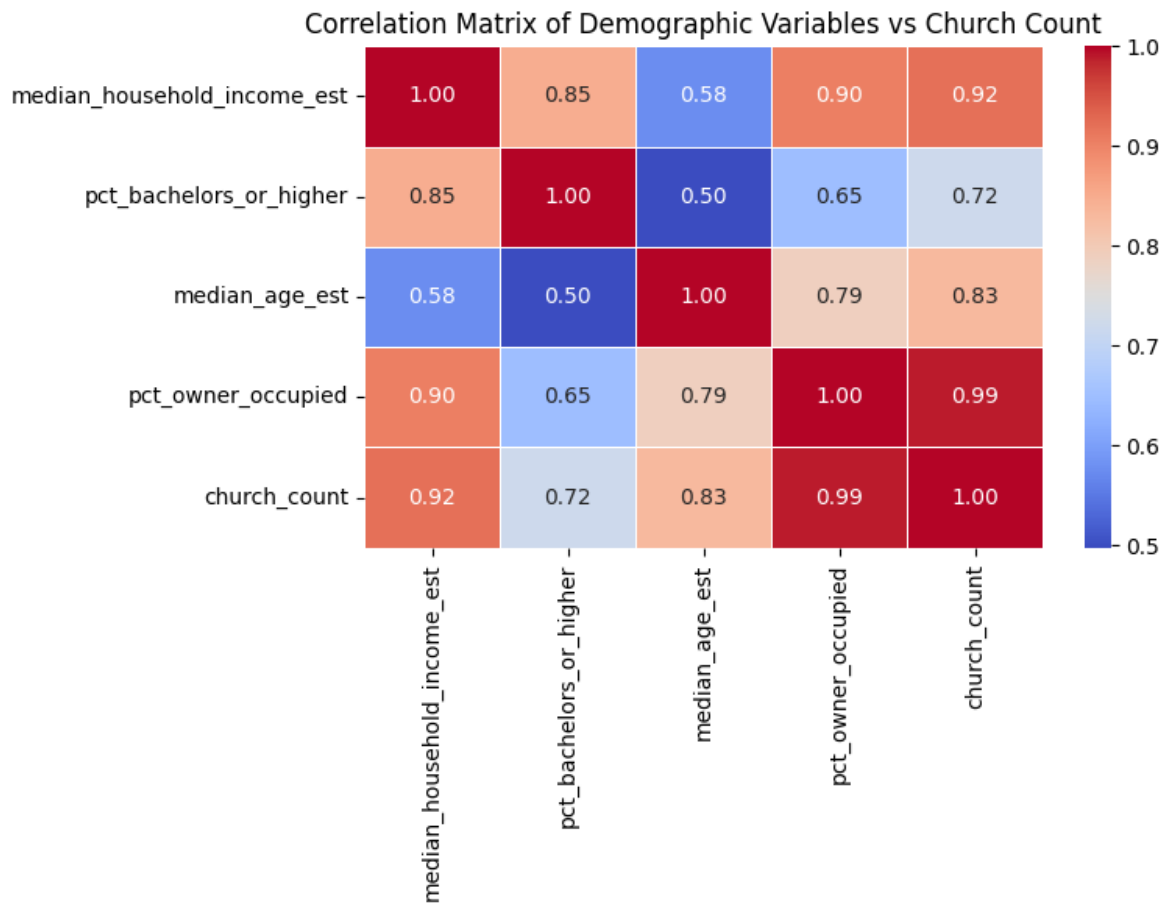
4. Data Overview

Radius	Population	% Two-Parent Households	% Single-Parent Households	Churches	Churches per 1000 People
1 mile	5,312	26.81%	73.19%	5	0.94
3 miles	41,041	34.77%	65.23%	64	1.56

Radius	Population	% Two-Parent Households	% Single-Parent Households	Churches	Churches per 1000 People
5 miles	98,220	41.18%	58.82%	88	0.90
7 miles	147,257	45.58%	54.42%	111	0.75
9 miles	180,527	48.68%	51.32%	121	0.67

Several trends are visible in the data:

- The proportion of **two-parent households increases with distance from the urban center**
- The proportion of **single-parent households decreases with distance**
- Total church count increases with geographic coverage
- **Church density per capita gradually declines as population growth outpaces church growth**



5. Correlation Analysis

A correlation matrix was calculated to evaluate relationships between demographic variables and church presence.

Variable	Correlation with Church Count
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Percent Owner Occupied	0.99
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Median Household Income	0.92
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Median Age	0.83
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Percent Bachelor's Degree	0.72
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These results suggest that church presence is most strongly associated with **stable residential environments**, particularly areas with higher homeownership rates and moderately higher incomes.

However, because population and institutional counts increase with geographic scale, these correlations should be interpreted cautiously.

6. Household Structure Trends

Household structure patterns provide additional context for understanding community composition.

Two-Parent Households

Radius Two-Parent Households

1 mile 26.81%

3 miles 34.77%

5 miles 41.18%

7 miles 45.58%

9 miles 48.68%

Two-parent households increase steadily as the radius expands, suggesting that **outer neighborhoods contain more traditional family structures**, which aligns with common suburban residential patterns.

Single-Parent Households

Radius Single-Parent Households

1 mile 73.19%

3 miles 65.23%

5 miles 58.82%

7 miles 54.42%

9 miles 51.32%

Single-parent households are more heavily concentrated near the urban core, reflecting typical demographic patterns observed in many U.S. cities.

7. Church Density Observations

Although the total number of churches increases with radius, **church density per capita gradually declines**.

Radius Churches per 1000 People

1 mile 0.94

3 miles 1.56

5 miles 0.90

7 miles 0.75

9 miles 0.67

This indicates that population growth occurs faster than church growth as geographic coverage expands.

8. Key Findings

Several key patterns emerge from the analysis:

1. **Homeownership shows the strongest correlation with church presence.**
2. **Median household income and church count are strongly correlated**, though this likely reflects neighborhood stability rather than income alone.
3. **Median age is positively associated with church presence**, consistent with national religious participation trends.
4. **Household structure varies significantly across geographic space**, with single-parent households more concentrated near the urban core.
5. **Church density decreases slightly as geographic scale expands**, indicating population growth exceeds church growth.

9. Implications for Local Ministry and Community Engagement

Examining the **1-mile radius surrounding the study location** reveals several notable characteristics.

Metric	1 Mile
Population	5,312
Two-Parent Households	26.8%
Single-Parent Households	73.2%
Owner Occupancy	24%
Churches	5

These figures suggest the immediate area has:

- Low homeownership
- High single-parent household concentration
- Potentially higher residential turnover
- Lower overall household stability

Such environments often experience higher economic stress and fewer stable social networks.

Importantly, the data indicates that **the challenge may not be a lack of churches**, as five churches already exist within one mile. Instead, the issue may involve **community stability and the capacity of residents to participate in structured institutions**.

10. Community Dynamics and Participation

Research in sociology and urban studies suggests that participation in institutions such as churches often declines in high-stress environments due to several factors:

Time constraints

Individuals working multiple jobs or managing single-parent households may have limited availability for regular participation.

Economic stress

Households facing financial insecurity often prioritize immediate needs such as housing, employment, and childcare.

Residential mobility

Higher renter populations often result in frequent relocation, weakening long-term community connections.

Fragmented social networks

Stable congregations often depend on multi-generational relationships that are more difficult to maintain in high-turnover neighborhoods.

These factors can make traditional church participation more difficult even when churches are physically present.

11. Strategic Implications for Churches

Historical patterns show that churches located in challenging environments often grow by **shifting their role from solely worship centers to community anchors.**

Successful approaches frequently include:

- childcare and family support programs
- youth mentorship and after-school activities
- food distribution and social services
- job training or community resource networks
- safe gathering spaces for local residents

In these cases, **community engagement often precedes congregational growth.**

12. Geographic Opportunity

The radius analysis also reveals a broader pattern.

As distance from the urban core increases:

- household stability improves
- two-parent household rates increase

- homeownership rises

This suggests a potential **dual strategy** for churches located in the urban core:

Local Mission

Serve the immediate community through outreach and social support.

Regional Congregation

Attract attendees from surrounding stable neighborhoods.

Many successful urban churches operate with this model.

13. Limitations

Several limitations should be considered when interpreting these results.

Small sample size

The dataset contains only five observations, limiting statistical inference.

Radius expansion bias

As geographic radius increases, population and institutional counts naturally increase, potentially inflating correlations.

Church presence vs attendance

The analysis measures church locations, not actual participation or membership.

OpenStreetMap completeness

OpenStreetMap may not capture every church within the study area.

14. Conclusion

This analysis demonstrates how **spatial demographic data can be used to explore the relationship between community characteristics and the distribution of social institutions**.

The findings suggest that church presence is most strongly associated with **stable residential environments**, particularly areas with higher homeownership rates, higher median incomes, and older populations.

At the same time, the data indicates that urban core areas may contain higher proportions of single-parent households and lower residential stability, which may influence participation in traditional institutions.

Rather than indicating decline, these patterns highlight the potential for churches to serve as **community anchors within neighborhoods facing greater social and economic challenges**.

